

GPU Programming

List of assignment problems

- Important Instructions:
- i) Attempt any ONE problem.
 - ii) Prepare a very small report containing your explanation/innovation and summary of results.
 - iii) Submit the code as a separate file (preferably NOT zip)
 - iv) If necessary, codes may be submitted as zip file.

Problem 1

Identify any problem that you are currently working on in your job role.

Identify any computational part (example-data manipulation pipelines) of the project. Abstract the problem so that the tasks could be done with random/open-source data. Implement this in CUDA.

Example tasks could be from areas such as Big Data Analytics, Graph Analytics, Financial Modeling (High-frequency Trading) or Recommendation Systems, etc.

Problem 2

Re-write kernels from an important application for any **one** of the following:

- a) AR/VR Simulation
- b) Protein folding/unfolding (Molecular Dynamics)
- c) Climate & Weather Modeling (Atmospheric and oceanic simulations)
- d) Real-time Rendering (Ray tracing, path tracing, lighting models)
- e) Fluid Dynamics (Turbulence modeling, aerospace design)

You can identify any library that already does the above, then write your own kernel and comment on performance of your kernel v/s library implementation over simplistic data samples (may not be real-life).

Problem 3

DL models are known to be computationally intensive during training. For example- Training GPT-4 or Stable Diffusion is impossible without GPU acceleration — CPUs would take **months or years** longer. However, inference is relatively easier. But, it remains computationally difficult for edge devices.

So, goal of this problem is: Parallel code development of inference of any complex model (GPT, BERT, LLaMA, ViT) could also be performed.

Problem 4

Implement a cryptography algorithm over GPU

Any one of the following two:

SHA / MD5 Hashing — compute digests in parallel for large datasets.

AES Encryption / Decryption — parallel block cipher operations.

Problem 5

Implement a partial differential equation (PDE) over GPU

A partial differential equation (PDE) is a mathematical equation that involves an unknown function of two or more variables and its partial derivatives. These equations are crucial for modeling natural phenomena and systems that change in both space and time, such as heat flow, sound waves, and fluid dynamics. A solution to a PDE is a function that satisfies the equation when it is substituted into it.

Example: Stencil-Based PDE Solvers — heat diffusion, Laplace, Poisson, or wave equations.

Problem 6

The goal of this assignment is to implement and optimize **matrix exponentiation** on a GPU. Please compare CPU and GPU implementations for computing a very large matrix power such as: A^{100} where A is a square matrix.

Problem 7

Implementation of the paper “Neon: A Multi-GPU Programming Model for Grid-based Computations” (paper uploaded on 16th May)

-Once this baseline is computed, as a bonus, attempt improvement over this.

Problem 8

Consider that you are tasked with development of Real-Time Tumor Detection from MRI/CT Images.

Use CUDA to accelerate tasks such as Image preprocessing, Segmentation and any DL model inference (example- CNN inference)

Problem 9

Your task is to convert 2D medical slices into 3D organ models using GPU rendering. This can be highly useful in applications such as medical surgery etc.